

Concept Review Questions

Directions for questions 1 to 30: For the Multiple Choice Questions, select the correct alternative from the given choices. For the Non-Multiple Choice Questions, write your answer in the box provided.

1. Thrice a number is 24 more than one-third of it. Find it.
2. Four added to half of one-third of one-sixth of a number is equal to one-twelfth of the number. What is the number?
(A) 48 (B) 36 (C) 72 (D) 24
3. The sum of two numbers is 18. The difference of the numbers is 4. Find the numbers.
(A) 11, 7 (B) 12, 8 (C) 16, 2 (D) 8, 10
4. How many pairs (x, y) satisfy the equations $3x + 6y = 12$ and $4x + 8y = 16$?
(A) 0 (B) 1 (C) 2 (D) ∞
5. How many pairs (x, y) satisfy the equations $3x + 9y = 21$ and $6x + 18y = 45$?
(A) 0 (B) 1 (C) 2 (D) ∞
6. How many pairs (x, y) satisfy the equations $3x + 5y = 21$ and $4x + 7y = 29$?
(A) 0 (B) 1 (C) 2 (D) ∞
7. If $8x + y = 10$ and $4x + 2y = 13$, then the respective values of x and y are
(A) 3, $4/3$ (B) $2/3$, $4/3$
(C) $3/4$, 2 (D) $7/12$, $16/3$
8. Find the value of x , if $3x + 9y + 12z = 18$ and $2x + 3y + 4z = 8$.
(A) 2 (B) 3 (C) 1 (D) 4
9. Five puffs and two samosas cost ₹48. The cost of a puff is twice the cost of a samosa. Find the cost of a puff (in ₹).
10. Four pens and five erasers cost ₹32. Five pens and four erasers cost ₹31. Find the cost of each pen (in ₹).
(A) 2 (B) 3 (C) 4 (D) 5
11. Four pens and three erasers cost ₹17. Four erasers and 3 pens cost ₹18. Find the total cost of a pen and a eraser.
12. The cost of ten mangoes and nine apples is ₹104. The cost of twenty-seven mangoes and twenty-five apples is ₹285. What is the cost of three mangoes and two apples?
(A) ₹24 (B) ₹30 (C) ₹27 (D) ₹32
13. Two chocolates, three milk shakes and four cakes cost ₹190. Four chocolates and eight cakes cost ₹320. Find the cost of a milkshake. (in ₹).
(A) 10 (B) 20
(C) 30 (D) Cannot be determined
14. A two digit number has a tens digit of T and a units digit of U. The digit 4 is placed immediately to the right of U forming a three digit number. The new number is equal to _____.
(A) $10T + U + 4$ (B) $100T + 10U + 4$
(C) $1000T + 10U + 4$ (D) $T + U + 4$
15. A two-digit number is such that twice the tens digit added to eleven times the units digit is equal to the number itself. Find the number.
16. The digits of a two digit number differ by 3. Find the difference of the number and the number formed by reversing its digits.
(A) 18 (B) 27 (C) 36 (D) 45
17. The difference between a three-digit number and the number formed by reversing its digits is 198. Find the difference of its first and last digits.
18. The greatest number that always divides the difference between a three digit number and the number formed by reversing its digits is
19. A fraction is such that the numerator is five less than the denominator. Also four times the numerator is one more than the denominator. Find the fraction.
(A) $4/9$ (B) $3/8$ (C) $2/7$ (D) $7/12$
20. The denominator of a fraction is two more than its numerator. If one is added to the denominator, the fraction becomes $3/4$. Find the fraction.
(A) $7/9$ (B) $5/7$
(C) $11/13$ (D) $9/11$
21. Three consecutive even integers are such that one-third of the second number is equal to one-fourth of the third number. Find the three numbers.
(A) 4, 6, 8 (B) 8, 10, 12
(C) 12, 14, 16 (D) 2, 4, 6
22. Nalini has an amount of ₹20 in coins of denominations of 50 paise and ₹1. If she has a total of 30 coins with her, how many ₹1 coins does she have?
23. Shankar has an amount of ₹7,000 in the denominations ₹500 and ₹100. How many ₹500 notes does he have, if he has a total of 22 notes with him?

24. Amar, Bhavan, Chetan and Dinesh have a total of ₹150 with them. Amar has one-fourth of the total amount with the others. Find the amount with Amar (in ₹).
(A) 20 (B) 25 (C) 30 (D) 37.5
25. P, Q and R had a total of ₹1250 among them. P had three-seventh of the total share with Q and R. Find the amount that P had (in ₹).
(A) 300 (B) 450 (C) 525 (D) 375
26. Twenty years from now, Sreedhar will be six times as old as he was twenty years ago. What is the present age of Sreedhar?
(A) 20 years (B) 24 years
(C) 28 years (D) 32 years
27. Ramesh is thrice as old as Suresh. Two years hence, Ramesh will be twice as old as Suresh. Find Ramesh's present age (in years).
(A) 2 (B) 3 (C) 4 (D) 6
28. Four years ago, Ashok's age, was four times that of Bala. Three years hence, Ashok's age will be thrice the age of Bala. Find Ashok's present age. (in years).
(A) 56 (B) 63 (C) 60 (D) 66
29. Mrudhula gave birth to twins, when she was twenty years old. After x years, the sum of the ages of Mrudhula and her twin children is 50 years. What is the value of x ?
30. Twenty seven years ago, a man's age was half his wife's present age. Sum of their present ages is 90 years. What are the respective present ages (in years) of the man and his wife?
(A) 48, 42 (B) 52, 38
(C) 54, 36 (D) 46, 44